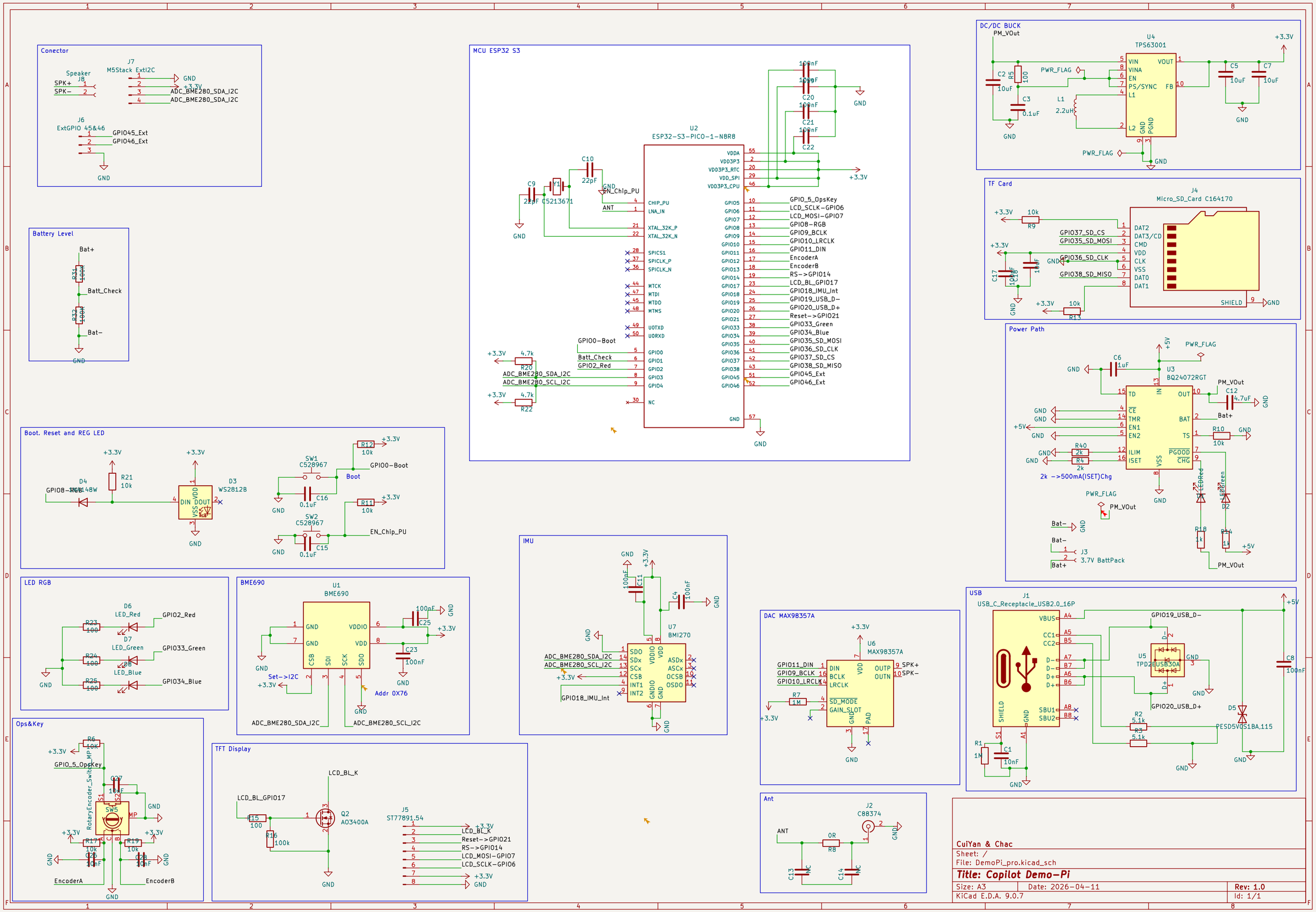
Conector
J7
M5Stack Ext12C
Speaker
J8
SPK+
SPK−
GND
+3.3V
ADC_BME280_SDA_I2C
ADC_BME280_SDA_I2C
J6
ExtGPIO 45&46
GPIO45_Ext
GPIO46_Ext
GND
Battery Level
Bat+
R31
100k
Batt_Check
R32
100k
Bat−
GND
Boot, Reset and REG LED
+3.3V
R21
10k
D4
WS2814BV
GPIO8−RGB
D3
WS2812B
DIN DOUT
VDD
VSS
GND
SW1
C528967
SW2
C528967
C16
0.1uF
C15
0.1uF
R12
10k
R11
10k
GPIO0−Boot
Boot
EN_Chip_PU
LED RGB
D6
LED_Red
GPIO2_Red
D7
LED_Green
GPIO33_Green
D8
LED_Blue
GPIO34_Blue
R23
100
R24
100
R25
100
BME690
U1
BME690
GND
VDDIO
VDD
CSB
SDI
SCK
SDO
C25
100nF
C23
100nF
Set−>I2C
Addr 0X76
ADC_BME280_SDA_I2C
ADC_BME280_SCL_I2C
Ops&Key
R6
10k
GPIO_5_OpsKey
RotaryEncoder_Switch_MP
SW5
C27
10nF
R17
10k
R19
10k
C26
10nF
C28
10nF
EncoderA
EncoderB
TFT Display
LCD_BL_K
LCD_BL_GPIO17
R15
100
R16
100k
Q2
AO3400A
J5
ST77891.54
LCD_BL_K
Reset−>GPIO21
RS−>GPIO14
LCD_MOSI−GPIO7
LCD_SCLK−GPIO6
MCU ESP32 S3
U2
ESP32−S3−PICO−1−N8R8
C10
22pF
C9
22pF
Y1
C5213671
EN_Chip_PU
ANT
XTAL_32K_P
XTAL_32K_N
SPICS1
SPICLK_P
SPICLK_N
MTCK
MTDI
MTDO
MTMS
U0TXD
U0RXD
GPIO0−Boot
Batt_Check
GPIO2_Red
ADC_BME280_SDA_I2C
ADC_BME280_SCL_I2C
R20
4.7k
R22
4.7k
NC
C19
100nF
C20
100nF
C21
100nF
C22
VDDA
VDD3P3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
GPIO_5_OpsKey
LCD_SCLK−GPIO6
LCD_MOSI−GPIO7
GPIO8−RGB
GPIO9_BCLK
GPIO10_LRCLK
GPIO11_DIN
EncoderA
EncoderB
RS−>GPIO14
LCD_BL_GPIO17
GPIO18_IMU_Int
GPIO19_USB_D−
GPIO20_USB_D+
Reset−>GPIO21
GPIO33_Green
GPIO34_Blue
GPIO35_SD_MOSI
GPIO36_SD_CLK
GPIO37_SD_CS
GPIO38_SD_MISO
GPIO45_Ext
GPIO46_Ext
IMU
U7
BMI270
C11
100nF
C4
100nF
SDO
SDx
SCx
CSB
INT1
INT2
GNDIO
ASDx
ASCx
OCSB
OSDO
ADC_BME280_SDA_I2C
ADC_BME280_SCL_I2C
GPIO18_IMU_Int
DAC MAX98357A
U6
MAX98357A
GPIO11_DIN
GPIO9_BCLK
GPIO10_LRCLK
DIN
BCLK
LRCLK
SD_MODE
GAIN_SLOT
OUTP
OUTN
SPK+
SPK−
R7
1M
Ant
J2
C88374
ANT
0R
R8
C13
NC
C14
NC
DC/DC BUCK
PM_VOut
U4
TPS63001
VIN
VINA
EN
PS/SYNC
L1
L2
VOUT
FB
GND
PGND
C2
10uF
R5
100
C3
0.1uF
L1
2.2uH
C5
10uF
C7
10uF
PWR_FLAG
TF Card
J4
Micro_SD_Card C164170
R9
10k
R13
10k
C17
10uF
C18
10uF
GPIO37_SD_CS
GPIO35_SD_MOSI
GPIO36_SD_CLK
GPIO38_SD_MISO
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
Power Path
U3
BQ24072RGT
C6
1uF
C12
4.7uF
R10
10k
R40
2k
R4
2k
2k −>500mA(ISET)Chg
TD
CE
TMR
EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
VSS
PM_VOut
Bat+
Bat−
J3
3.7V BattPack
LED_Red
LED_Green
D1
D2
R18
1k
R14
1k
USB
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
GND
U5
TPD2EUSB30A
GPIO19_USB_D−
GPIO20_USB_D+
R2
5.1k
R3
5.1k
R1
1M
C1
10nF
C8
100nF
D5
PESD5V0S1BA,115
CuiYan & Chac
Sheet: /
File: DemoPi_pro.kicad_sch
Title: Copilot Demo−Pi
Size: A3
Date: 2026−04−11
Rev: 1.0
KiCad E.D.A. 9.0.7
Id: 1/1